Starting Finance Club PoliTo

Research Paper

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RESEARCH ARTICLE

Operating-Value Sensitivity for Barrick Mining

Option-implied gold and copper laws in a controlled operating experiment

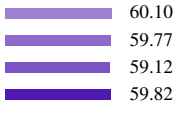
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JEL classification
G12; G13; G31; C32; C53; C61; C63

Discoveries

Calibrated engines share one unreconciled operating contract.



Signed-terminal proxy; USD billion. Sensitivity only; not company value or a physical probability forecast.

Abstract

This study audits the sensitivity of an unreconciled Barrick operating-value proxy to alternative option-implied commodity-price laws. Eight student teams connect probability, econometrics, simulation, operating forecasts and a common discounted-margin mapping. Black–Scholes, Heston, Bates and Bates–Hawkes are fitted to a common GLD surface. A calibration audit retains feasible nested candidates; historical gold out-of-sample scores remain a separate, frozen experiment. A new copper extension uses American options on matched COMEX futures, historical snapshots and both rolling and fixed-origin repricing tests. Three producing copper mines enter on Barrick's attributable sales basis. Production, unit costs and discount-rate shocks are controlled across paired gold-only and gold-plus-copper simulations. The copper contribution is reported primarily over five years without terminal continuation; its fixed-volume operating scenario does not forecast mine expansion. Aggregate operating proxies do not assert a net-debt, ownership or diluted-share bridge. Option-implied coefficients transferred to an artificial scenario law and combined with WACC do not constitute a measure change, corporate pricing or a physical forecast. Persistence benchmarks, numerical sensitivity and terminal conventions constrain interpretation of model rankings. The contribution is a reproducible research experiment with explicit limits on accounting, probability measure, exercise conventions, extrapolation and statistical inference. Results are neither fair values nor price targets, investment recommendations or evidence of market mispricing.

RESEARCH PROGRAMME AND COLLABORATION

This paper grew out of an eight-team project involving 17 students with complementary backgrounds in mathematics, data science, engineering, economics and finance. The size of the author group reflects a division of research responsibilities and a shared learning process: Teams 1–3 developed the mathematical, econometric and simulation foundations; Teams 4–5 built the operating and corporate-valuation layers; and Teams 6–8 studied gold-price dynamics, market evidence and option-implied calibration.

The extended technical report records definitions, derivations and implementation choices to establish a common technical language across these backgrounds. Teams worked on dedicated modules, then compared assumptions, revised their contributions and reconciled dates, units and interfaces before integration. The present paper condenses this work into a single auditable experiment. Together with the reproducible code, it demonstrates the group's existing research practice and provides a basis for developing a sustained student-led quantitative research team with academic guidance.

1 INTRODUCTION

A gold producer's value depends on the distribution of future commodity prices as well as its operating and financing assumptions. A single price path can hide this source of model risk. We ask how an unconstrained Barrick operating-proxy distribution changes when progressively richer gold-price processes are substituted within a common operating scenario and discounted-margin contract.

We compare Black–Scholes, Heston, Bates–Poisson and Full Bates–Hawkes calibrated to GLD options. Current-surface diagnostics and rolling out-of-sample tests distinguish fit from predictive performance; shared production, costs and WACC shocks isolate the effect of the gold engine. The contribution is this controlled comparison and its auditable integration, rather than a claim that any model provides an unconditional fair value.

2 DATA AND VALUATION SETUP

2.1 Controlled layer separation

The integrated experiment is organized as a controlled handoff across three modules. Team 8 governs only the gold-price process; Team 4 supplies the operating forecasts; and Team 5 maps those inputs through a common tax, WACC, terminal-value contract without a reconciled equity bridge. In the four-model valuation comparison, only the gold engine changes. Barrick's Q1–Q2 2026 actuals and the same Team 4 production and unit-cost forecasts from Q3 2026 onward are frozen over 20 quarters; the same Team 5 corporate rules and WACC shocks are then reused across runs.

2.2 Calibration snapshot and measure discipline

The current calibration uses the 2 September 2026 GLD call surface acquired through the IB API. After the common DTE ≥ 75 and numerical filters, the dense surface contains 605 eligible observations across 12 expiries and 146 distinct strikes, with 79–653 days to expiry. The official 8×8 Chebyshev–Chebyshev sample contains 64 distinct actual contracts spanning eight expiries and 20 strikes. The risk-free input is a same-date NSS fit to 14 U.S. Treasury tenors (Svensson, 1994; U.S. Treasury, 2026). Its 2.0608 bp RMSE is computed against the continuously compounded par-yield proxy; the curve is not described as a bootstrapped zero curve. The Barrick market reference is the 2 September close of USD 44.13, while the common gold anchor remains Barrick's Q2 2026 average realized price of USD 4,417/oz (Barrick, 2026). The operating calendar is a fixed scenario index, not a forecast commencing on 2 September; earlier actuals are not future observations.

The option layer is calibrated under $\mathbb{Q}$, with deterministic pricing rates. The operating experiment instead defines an artificial scenario law $\mathcal{R}$: calibrated coefficients are reanchored, commodity mean schedules imposed and independent WACC shocks sampled. Neither an estimated Radon–Nikodym density nor validated physical risk premia enter this frozen experiment. Consequently $\mathcal{R}$ is neither a validated $\mathbb{P}$ forecast nor the original traded-instrument $\mathbb{Q}$ law. Gold uses GLD shape

at a separately dated realised-gold level; copper uses a cross-delivery mean schedule. Neither transformation is a change of measure.

For a fixed-delivery futures price, the deterministic-rate pricing convention has zero drift. Copper's delivery-curve slope is a scenario assumption, not that futures time drift. Jump compensation uses the simulated intensity and mark moment $\lambda(\mathbb{E}[e^J]-1)$. A genuine $\mathbb{Q}\to\mathbb{P}$ specification would need price and variance risk premia and, with jumps, intensity and mark-law adjustments; changing drift alone would not estimate them. Futures are also not generally physical expected spot prices (Schwartz, 1997).

Ordinary DCF uses physical expected cash flows with a compatible risk-adjusted discount rate; a pricing-measure valuation instead needs its consistent numeraire and pricing kernel. Q-shaped scenarios plus assumed WACC may mix incompatible risk adjustments. Replacing WACC by a risk-free curve would not by itself reconcile this incomplete corporate claim. WACC shocks are not short-rate shocks. The reported operator is therefore a conditional sensitivity, without a corporate-value or real-world probability interpretation.

Physical forecasting interface. Section 3.4 specifies the cross-measure link, new historical inputs and initial physical validation. These diagnostics remain separate from the frozen scenario operator: no estimated density or validated physical risk premium is used in its reported values.

Table 1. Frozen contract used for model attribution.

Block	Current experiment
Gold layer	Black–Scholes/GBM, Heston, Bates–Poisson and Full Bates–Hawkes calibrated by Team 8 under $\mathbb{Q}$.
Operations	Barrick Q1–Q2 2026 actuals followed by Team 4 deterministic production and unit-cost vectors from Q3 2026.
Corporate mapping	Team 5 tax, growth–ROIC–reinvestment schedule, stochastic WACC and explicit terminal convention; no equity bridge.
Simulation	8,192 paths, 260 fine steps, 20 quarters and a common WACC shock array.

2.3 Option exercise convention

GLD contracts are American-style (Cboe, 2026). We set payout yield $q=0$: trust expenses reduce backing, not cash distributions (SPDR, 2025). For non-distributing calls and positive rates, early exercise is suboptimal in the frictionless model. An 800-step CRR check on all 64 contracts gives an American–European premium below 10^{-8} USD; this tests that convention, not market frictions.

2.4 Operating-model status

Team 4's original operating study is richer than the object ultimately passed downstream. Cost of Sales is forecast from a log-space master chain; production is constructed bottom-up from ore processed, average grade and recovery. The unified valuation refreshes the handoff with 2026 actuals but still freezes the resulting 20-quarter production and cost vectors. Consequently, mine-level operating uncertainty is not propagated jointly with gold-price uncertainty. Team 4's legacy price simulation and illustrative EBITDA/valuation output are explicitly excluded.

2.5 Gold perimeter and reporting basis

The Team 4 subset comprises Bulyanhulu, Carlin, Cortez, Kibali, Loulo–Gounkoto, North Mara, Pueblo Viejo and Turquoise Ridge. Its quarterly estimation window is 2019Q1–2025Q4, with North Mara beginning in 2019Q4. The operating quantities are already attributable: Nevada Gold Mines is 61.5%, Pueblo Viejo 60%, Kibali 45%, Loulo–Gounkoto 80%, and the Tanzanian mines 84%. Kibali is equity-accounted, whereas the other admitted interests are consolidated with partners' non-controlling interests (Barrick, 2025). Those partners must not be deducted again from attributable flows. Tanzania's contractual economic-benefit sharing still requires a separate cash-flow reconciliation.

The frozen handoff mixes company-wide gold actuals of 719 and 796 koz in Q1–Q2 2026 with later forecasts for the eight-mine subset; that subset produced only 622 and 711 koz in those quarters. The scope break is retained and disclosed, not silently described as a consistent mine panel. Cost of Sales includes depreciation and price-sensitive royalties; multiplying its sold-ounce cost by production yields a component-margin proxy, not EBITDA or reconciled EBIT. Loulo–Gounkoto's 2025Q4 accounting uplift also makes its cost anchor a normalization issue. Ownership, the North America IPO, cash/debt, share count and reserves must be frozen at a common corporate valuation date before a per-share result is permitted.

3 METHODOLOGY

3.1 Gold-price engines

The four Team 8 specifications relax one restriction at a time. Under $\mathbb{Q}$, Black–Scholes/GBM uses constant volatility,

$$\frac{\mathrm{d}S_t}{S_t} = (r_t - q_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t. \tag{1}$$

GBM is the constant-volatility benchmark. Heston introduces mean-reverting stochastic variance (Heston, 1993),

$$\frac{\mathrm{d}S_t}{S_t} = (r_t - q_t)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S, \tag{2}$$

$$\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v, \qquad \mathrm{d}W_t^S\,\mathrm{d}W_t^v = \rho\,\mathrm{d}t. \tag{3}$$

Bates adds compensated compound-Poisson jumps (Bates, 1996),

$$\frac{\mathrm{d}S_t}{S_{t^-}} = (r_t - q_t - \lambda k)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S + (e^J - 1)\,\mathrm{d}N_t, \tag{4}$$

where $k = \mathbb{E}^{\mathbb{Q}}[e^J - 1]$.

3.2 Full Bates–Hawkes: implemented affine engine

The full engine preserves Heston variance but replaces the constant Poisson arrival rate with an exponential Hawkes state (Hawkes, 1971):

$$\frac{\mathrm{d}S_t}{S_{t^-}} = (r_t - q_t - \lambda_t k)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S + (e^J - 1)\,\mathrm{d}N_t, \tag{5}$$

$$\mathrm{d}v_t = \kappa(\theta_v - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v, \tag{6}$$

$$\mathrm{d}\lambda_t = \beta(\bar\lambda - \lambda_t)\,\mathrm{d}t + \alpha\,\mathrm{d}N_t, \qquad n = \alpha/\beta < 1. \tag{7}$$

Under the implemented exponential-kernel specification, pricing exploits the affine Markov structure rather than a standalone finite-difference PIDE. Writing $X_t = \log S_t$, the characteristic function factorizes into Heston diffusion and Hawkes jump blocks,

$$\phi_{X_T}(u) = \phi_{\mathrm{diff}}(u)\exp(A(u,T) + B(u,T)\lambda_0)\,, \tag{8}$$

with the jump block determined by the Riccati system

$$\frac{\mathrm{d}B}{\mathrm{d}\tau} = \psi_J(u)e^{\alpha B} - \beta B - iuk - 1, \tag{9}$$

$$\frac{\mathrm{d}A}{\mathrm{d}\tau} = \beta\bar\lambda B, \qquad A(0) = B(0) = 0, \tag{10}$$

where $\psi_J(u) = \exp(iu\mu_J - \frac{1}{2}\sigma_J^2u^2)$. European call prices use Fourier-COS inversion (Fang and Oosterlee, 2008), with independent diffusion and jump drivers. The identities $\phi(0)=1$, the martingale condition at $u=-i$, and the limit $\alpha\to 0$, $\lambda_0=\bar\lambda$, back to Bates are used as implementation checks.

Model-role constraint

These processes govern commodity prices only. The copper extension uses forward-price analogues; neither price layer generates Barrick production or costs.

3.3 Option-surface calibration and predictive test

Every model is fitted to the same structured GLD surface by minimizing a vega-scaled price loss,

$$F(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{C_i^{model}(\theta) - C_i^{mkt}}{\max(\mathrm{Vega}_i, \varepsilon)}\right]^2, \tag{11}$$

with Vega per unit decimal volatility and $\varepsilon = 10^{-4}$. This is price-based MSE; exact IV RMSE uses inversion, a square root and 10^4 scaling.

Nested starts are retained during SLSQP refinement. A Feller strict-positivity constraint is imposed by choice; Hawkes stability requires $n < 1$.

Calibration fit and forecasting are evaluated separately. A date is admitted to the rolling test only when at least 64 eligible observations and three expiries survive the common filters. Each CC64 origin is scored on the next dense date. With n_t common targets on date t, equal-date performance is

$$R^2_{OOS,m|b} = 1 - \frac{\sum_t n_t^{-1} \sum_j e^2_{m,t,j}}{\sum_t n_t^{-1} \sum_j e^2_{b,t,j}}, \tag{12}$$

against both the origin cross-sectional mean IV and an origin observed-IV surface interpolated in maturity and log-moneyness inside its convex hull. The target spot and rate curve enter only as scoring-date conditioning variables. This design prevents the lowest in-sample IV error from being interpreted automatically as predictive dominance.

Origin parameters are frozen before next-date scoring; target quotes do not select them. Conditioning on target spot and rates makes this a repricing test, not a joint ex ante forecast. Persistence is evaluated only on its interpolation support; unmatched observations are not zero losses.

3.4 From pricing measure to physical revenues

Option coefficients under $\mathbb{Q}$ do not identify physical commodity returns. All four families can be estimated under $\mathbb{P}$, but their cross-measure link requires underlying history and identified risk premia (Pan, 2002). With $\mathrm{d}W^Q = \mathrm{d}W^P + \eta\sqrt{v}\,\mathrm{d}t$, the restricted variance map is $\kappa_P = \kappa_Q - \xi\eta_v$ and $\kappa_P\theta_P = \kappa_Q\theta_Q$. A positive intensity scaling c_I and normal mark tilt give $\lambda_P = c_I\lambda_Q$ and $\mu_{J,P} = \mu_{J,Q} + b_J\sigma_J^2$. Thus physical expected price return is

$$\mu_t^P = a_t^Q + \eta_S v_t + \lambda_t^Q(c_I k_P - k_Q), \tag{13}$$

where a_t^Q is pricing carry and $k_M = \mathbb{E}^M[e^J - 1]$. Hawkes baseline, initial intensity and excitation scale by c_I, while decay is unchanged; physical branching must stay below one. Algebraic closure does not identify premia. For the restricted CIR and unmarked linear Hawkes family, a stopped-density entropy bound establishes equivalence on finite horizons; its full assumptions and derivation are in the unified study (Schönbucher, 2001; Sokol and Hansen, 2015). Empirical estimation and predictive validation remain separate. In particular, copper Bates has zero fitted Q jump intensity: an equivalent change cannot create physical jumps without a new joint specification.

The 6 October diagnostic, still cut off at 2 September, adds 1,253 GLD regular-session prices and 1,003 synchronous GLD–gold observations, including early closes. Seven copper deliveries provide 408 accepted origin-selected, same-delivery returns; four requested earlier deliveries were unavailable. This avoids artificial rollover jumps but does not identify copper spot carry or realised selling prices. The approximately 100-contract option budget is separate from the daily time-series sample.

A training-only GLD GBM with an origin-fitted gold basis produces quarterly gold-price RMSE of USD 417.19/oz over 689 overlapping targets, versus USD 428.52/oz for a no-change mean. Its nominal 95% coverage is only 89.6%; direct gold GBM has better absolute error and log score. This is a physical historical benchmark, not a joint option/history fit. Preliminary fixed-current-Q Heston, Bates and Hawkes premium estimates remain sensitive to particle and time-step resolution and are not admitted to valuation.

Option repricing OOS, physical commodity OOS and Barrick operating OOS are distinct tests. Current pricing coefficients cannot be used at earlier physical forecast origins. The intended DCF requires $\mathbb{P}$ realised selling prices. Thirty recovered historical surfaces, each with 64 contracts, permit new dated calibrations. A conditional two-stage estimator may fit $\mathbb{Q}$ options and then physical premia using origin-available returns; it is not joint maximum likelihood and must address uncertainty from both stages. Forward paths share the option-estimated initial state across measures. All four physical models now have 29 common next-session targets. After origin-only gold conversion, RMSE is USD 66.35/oz for GBM, 66.77 for Heston, 66.84 for Bates and 65.81 for Bates–Hawkes, versus 66.09 for no-change. The small Hawkes advantage does not establish superiority: nominal 95% coverage is 27/29 for GBM/Heston and 25/29 for both jump models, with parameter and basis uncertainty omitted. Two Heston fits and one Hawkes fit fail to converge and remain included; independent filter checks show substantial seed sensitivity. Resumed fits use strictly earlier parameters as initial guesses with global fallback, while preserving prior cold fits and recording the initialization date/hash. This short panel does not validate long-horizon physical distributions. Analytically integrating normal jump marks reduces seed dispersion in a targeted frozen-parameter check, but substantial sensitivity remains; the reported panel uses the original filter. A fixed-delivery copper future cannot be extended beyond expiry as a spot forecast, and a September valuation must exclude elapsed quarterly cash flows. The operating bridge also requires attributable sales, reconciled FCFF and compatible discounting. WACC does not convert $\mathbb{Q}$ paths to $\mathbb{P}$; frozen scenario values below remain a separate experiment. The unified study gives the complete coefficient derivation, data exclusions, numerical checks and daily/quarterly baseline results.

3.5 Team 4 cost and production models

For costs, Team 4 first builds a log-space global master chain from mine-level series and then selects an **ARIMA(3,1,0) with drift** by automated information-criterion search. The admitted specification is

$$(1 + 0.9501B + 0.6128B^2 + 0.3003B^3)(1 - B)y_t = 0.0222 + \varepsilon_t. \tag{14}$$

The selected model minimizes both $AIC = -94.64$ and $BIC = -86.87$ (Akaike, 1974; Schwarz, 1978), where p counts parameters and N observations,

$$AIC = 2p - 2\log\widehat{L}, \qquad BIC = p\log N - 2\log\widehat{L}. \tag{15}$$

The source's 12-quarter cost hold-out reports log-scale RMSE of 0.0504 against 0.1512 for a random walk. Its 0.603% MAPE is computed on log values, not USD/oz; it must not be read as a cost-level percentage error. This is source evidence, not independent replication.

Team 4 models production components without claiming AIC/BIC selection. With ore in kt, grade in g/t and recovery as a fraction, physical output in koz is

$$Q_t^{phys} = \mathrm{Ore}_t\,\mathrm{Grade}_t\,\mathrm{Recovery}_t/31.1035, \tag{16}$$

with ore modelled as $SARIMA(1,0,1)(1,0,0)_4$, grade as a non-seasonal AR(1), and recovery sampled uniformly from historical extrema widened by one percentage point and truncated to $[0.50, 1.00]$. Team 4 additionally applies a 0.91 empirical realization factor, not a unit conversion; its transfer across mines remains unvalidated. The valuation retains the frozen Team 4 vectors, not the Goodman distributions; production bands are 80% and cost bands 95%, while product-variance formulas require explicit dependence assumptions.

3.6 Valuation mapping

Team 5 makes growth internally consistent with reinvestment through

$$b_t = \frac{g_t}{ROIC_t}, \qquad F_t^{proxy} = M_t^{AT} - \max(M_t^{AT}, 0)b_t, \tag{17}$$

where M_t^{AT} is the after-tax component margin. Over the five explicit years, g_t declines linearly from 8.50% to 2.50%, $ROIC_t$ from 14.50% to 10.50%, and the implied reinvestment rate from 58.62% to 23.81%.

The common discount-rate process is mean reverting,

$$w_{t+1} = \bar{w} + (w_t - \bar{w})e^{-\kappa_w} + \sigma_w^*\varepsilon_{t+1}, \tag{18}$$

with $w_0 = 9.10\%$, $\bar{w} = 8.30\%$, $\kappa_w = 0.45$ and annual volatility 0.90%. It is clipped to $[g_\infty + 1.00\%, 25.00\%] = [3.50\%, 25.00\%]$; every gold engine receives the same 8,192-path shock array.

For simulated path i, the common operating contract and discount rates define the operating proxy,

$$V_{op}^{(i)} = \sum_{t=1}^{T} \frac{F_t^{proxy,(i)}}{\prod_{s=1}^{t}(1 + WACC_s^{(i)})} + \frac{TV_T^{(i)}}{\prod_{s=1}^{T}(1 + WACC_s^{(i)})}. \tag{19}$$

The terminal block uses $g_\infty = 2.50\%$, $ROIC_\infty = 10.50\%$ and therefore $b_\infty = 23.81\%$:

$$TV_T^{(i)} = \frac{M_T^{AT,(i)}(1+g_\infty)(1-g_\infty/ROIC_\infty)}{WACC_T^{(i)} - g_\infty}, \tag{20}$$

The WACC floor ensures a positive spread. A signed terminal removes the legacy zero floor; $TV = 0$ gives a five-year-only sensitivity. Neither closure is a reserve-calibrated mine life or an optimal abandonment rule. No equity/share bridge is inferred.

4 CURRENT OPERATING-MODEL EVIDENCE

Team 4 supplies the inherited operating scenarios, refreshed with Q1–Q2 2026 statistics. Its earlier gold simulation is excluded: all price paths come from Team 8. These fixed inputs permit model attribution but are not independently validated operating forecasts.

4.1 Cost-of-sales block

The cost model in Section 3.5 extracts a common log-space trend, forecasts it with ARIMA(3,1,0) with drift, and reconstructs mine-level paths with volatility-scaled anchors. Its reported selection and hold-out diagnostics are inherited evidence, not revalidated forecasts or a jointly simulated cost state.

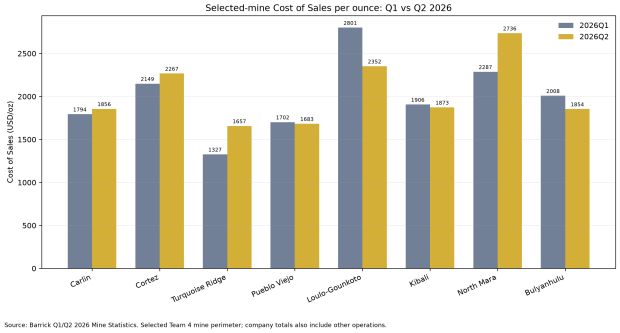


Figure 1. Selected-mine Cost of Sales per ounce in Q1 and Q2 2026. Mine dispersion motivates preserving the Team 4 operating decomposition rather than replacing it with a single company-average cost.

The scenario uses 20 fixed quarterly cost entries, including two 2026 actuals and 18 forecast entries. It does not propagate the source study's operating distributions.

4.2 Production block

Production is reconstructed from ore processed, grade and recovery using the specifications in Section 3.5. This preserves mine heterogeneity. The SARIMA orders are admitted from the source study, which does not report their selection by AIC/BIC.

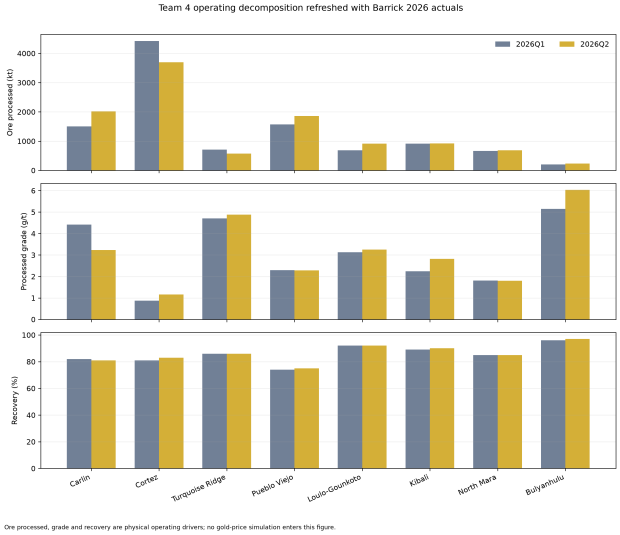


Figure 2. Ore processed, grade and recovery updated with Barrick 2026 actuals. These are physical operating drivers; no commodity-price simulation enters the panel.

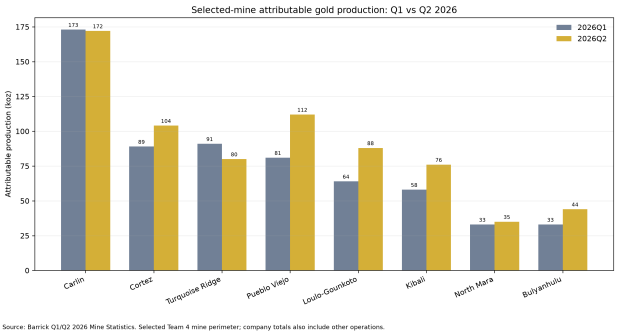


Figure 3. Selected-mine attributable gold production in Q1 and Q2 2026. The current actuals reveal heterogeneous changes that are hidden by a company-level total.

The original study also develops Goodman variance propagation and confidence bands for the production components. Those distributions remain useful evidence on operating-model uncertainty, but they are not propagated jointly in the four-engine corporate experiment: only the refreshed point vectors are passed downstream.

4.3 Forecast handoff used by the valuation

The final handoff keeps observed and modelled quantities distinct. Barrick's Q1–Q2 2026 actual production and CoS establish the current operating state; the Team 4 forecasts begin at Q3 2026 and extend through the common valuation horizon.

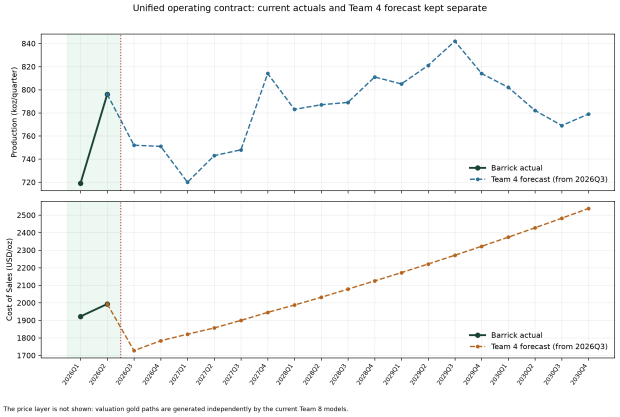


Figure 4. Inherited operating contract. Company-wide Q1–Q2 2026 actuals precede the eight-mine Team 4 forecast from Q3; the scope break is disclosed in Section 2.5. The straight cost path is the conditional mean of a persistent-trend specification, not a claim that operating uncertainty disappears.

Prudential operating boundary

Option-implied laws supply conditional scenario shapes to the proxy layer. Costs retain a persistent trend without additional technology-driven cost reductions. This is prudential relative to otherwise identical efficiency-improvement scenarios, but it does not establish a mathematical lower bound. Stochastic costs could yield favourable and adverse outcomes; the current fixed vectors preserve attribution across gold engines. Growth is already included in the common DCF schedule.

5 CALIBRATION SAMPLE AUDIT

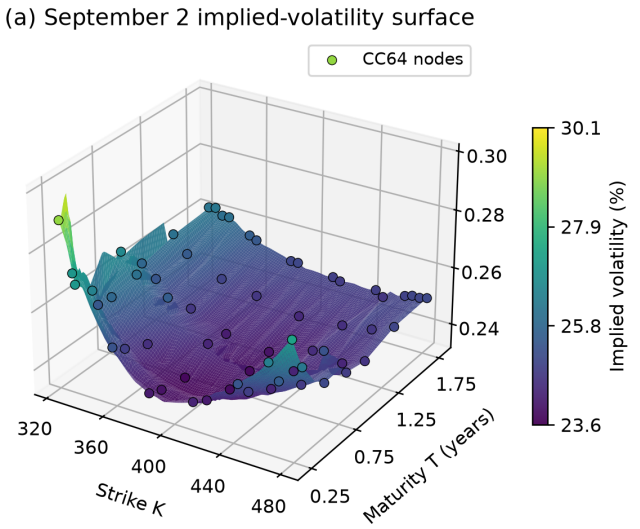


Figure 5. (a) GLD implied-volatility surface, 2 September 2026, with CC64 nodes. Interpolation is used for display only.

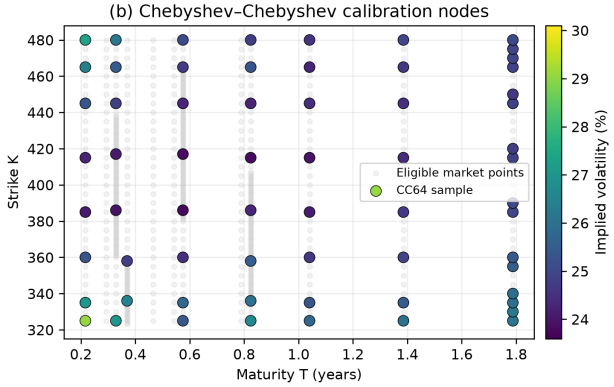


Figure 5 (continued). (b) The 64 actual contracts selected from 605 eligible calls across 12 expiries and 146 strikes.

5.1 Why structured sampling is retained

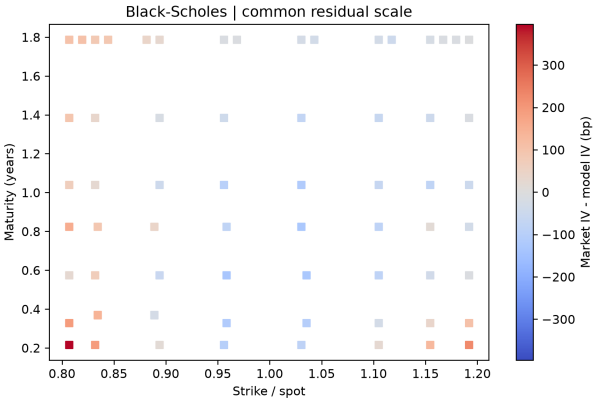
The option surface is irregular in strike and maturity. Team 8 therefore passes a fixed 8×8 Chebyshev–Chebyshev subset to the optimizer rather than allowing densely listed regions to dominate the loss. Every target is mapped to the nearest still-unused actual option: the 64-point calibration set contains no synthetic contracts. The figure audits the same 2 September parameter snapshot used by the refreshed valuation.

Evidence status

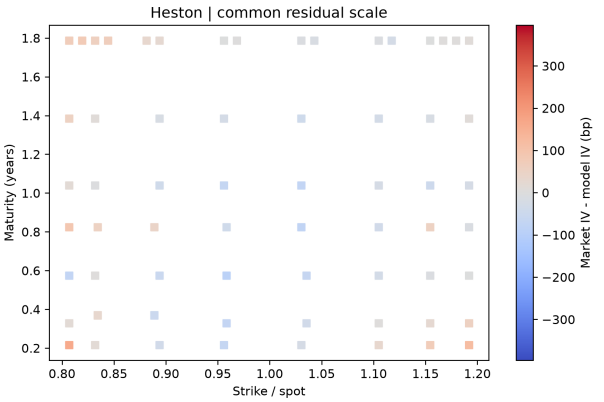
The calibrated inputs, residual maps and valuation paths share the same 2 September Team 8 parameter set. The U.S. Treasury curve is dated 2 September as well; the separate Q2 realized-gold level anchor remains explicit in the valuation contract.

5.2 What the residual maps add

The residual heatmaps show where each specification systematically over- or under-prices the observed implied-volatility surface. Black–Scholes leaves the strongest smile and term-structure pattern. Table 2 compares current-surface fit; Audited Bates and Hawkes fits improve on the nested starts. Hawkes branching is 0.5594. All maps use one colour scale; calibration fit does not establish economic superiority.

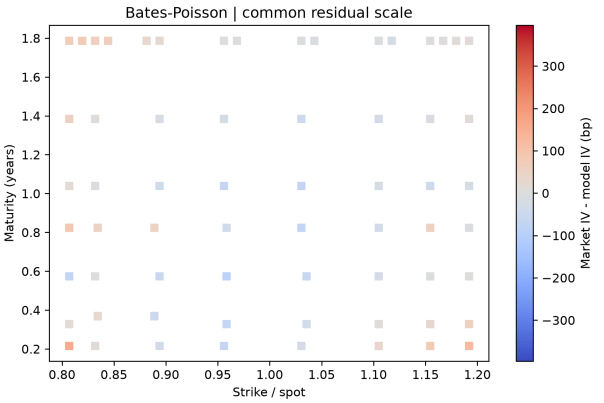


(a) Black–Scholes

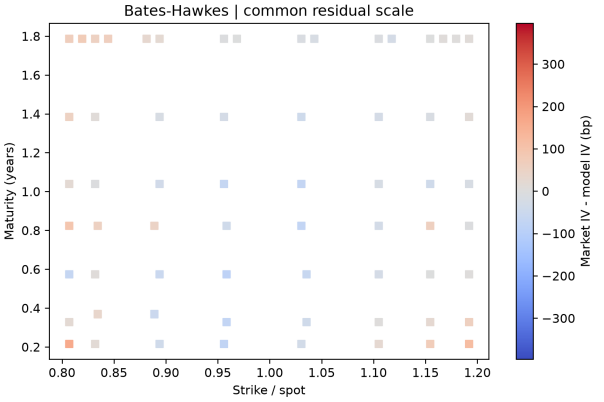


(b) Heston

Figure 6. Implied-volatility residual maps for the common 2 September CC64 sample. Blue denotes model IV above market IV and red denotes model IV below market; the audited fits retain feasible nested candidates.



(c) Bates–Poisson



(d) Full Bates–Hawkes

Figure 6 (continued). Bates–Poisson and Full Bates–Hawkes residual maps; all colourbars share the same residual range.

Table 2. Current-surface IV calibration RMSE.

Model	RMSE (bp)
Black–Scholes	97.2107
Heston	49.4881
Bates–Poisson	49.4077
Full Bates–Hawkes	49.3258

6 DCF DRIVER AND TERMINAL-VALUE AUDIT

Growth, reinvestment, WACC and terminal closure are shared across gold engines. Table 3 summarizes the corporate assumptions.

6.1 Growth, ROIC and reinvestment

Growth and ROIC converge linearly to their stable levels over five years. Reinvestment follows $b_t = g_t / ROIC_t$, linking growth to the capital required to sustain it (Figure 7).

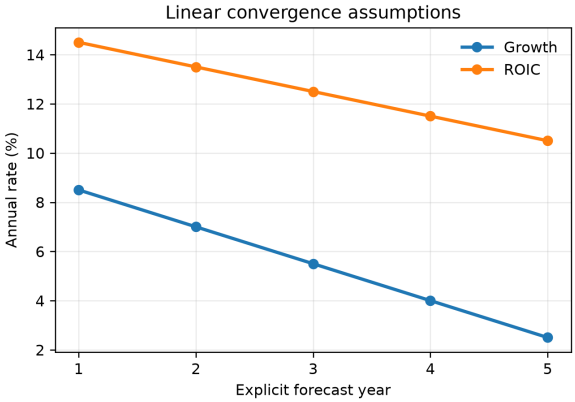


Figure 7. Common explicit-period DCF schedule: growth and ROIC converge linearly over the five-year horizon.

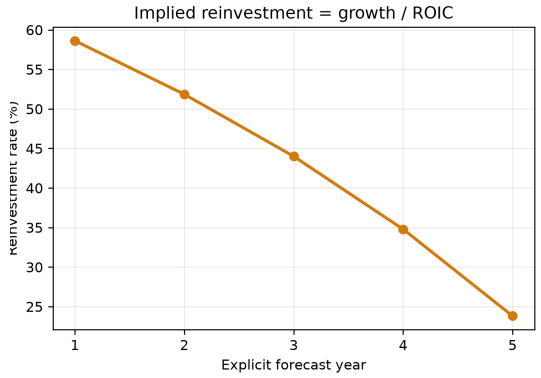


Figure 7 (continued). The implied reinvestment rate, defined as growth divided by ROIC, falls from 58.62% to 23.81%.

6.2 Common stochastic WACC

WACC follows the bounded mean-reverting process in Equation (17), with inputs in Table 3. Its annual innovation scale is $\sigma_w^* = \sigma_w \sqrt{(1 - e^{-2\kappa_w})/(2\kappa_w)} = 0.7308$ percentage points before clipping. All engines reuse 8,192 annual shock paths (Figure 8).

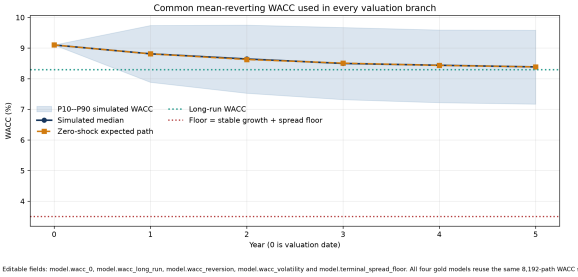


Figure 8. Common mean-reverting WACC. The band is the P10–P90 range from the shared 8,192-path shock array; the orange line is the zero-shock path and the dotted lines show the long-run level and admissibility floor.

6.3 Terminal closure and sensitivity

The terminal block uses $g_\infty = 2.50\%$, $ROIC_\infty = 10.50\%$ and terminal reinvestment of 23.81%. At the current median terminal WACC of 8.384%, the normalized terminal multiple is 13.27× final after-tax margin. The spread floor enforces $WACC_T > g_\infty$. No simulated rate reaches either clipping bound in the base run.

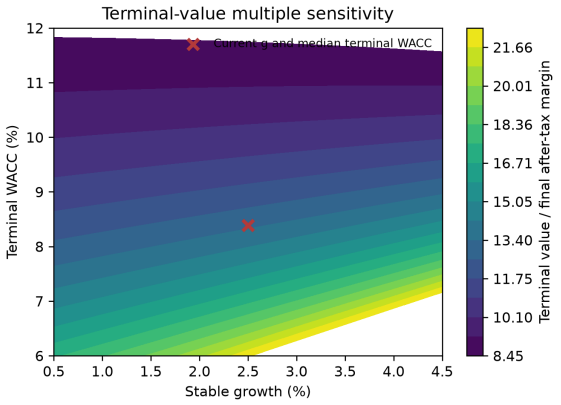


Figure 9. Terminal-value multiple sensitivity to stable growth and terminal WACC. The cross marks the configured stable-growth rate and current median terminal WACC.

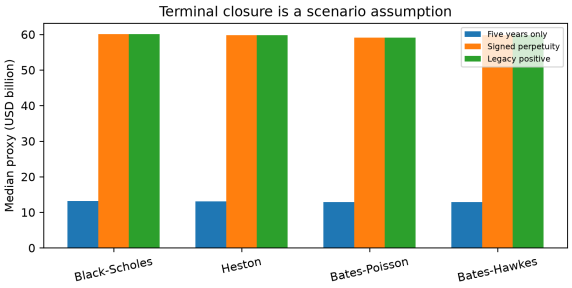


Figure 9 (continued). Median proxy under three terminal conventions. Five-year-only values are about USD 13 billion versus about USD 60 billion with perpetuity. Zero terminal is a truncation scenario, not an estimated reserve life.

Table 3. Editable common DCF inputs.

Parameter	Value
Tax rate	25.00%
$g_1 \to g_\infty$	8.50–2.50%
$ROIC_1 \to ROIC_\infty$	14.50–10.50%
$WACC_0 \to \bar{w}$	9.10–8.30%
WACC reversion	0.45
WACC volatility	0.90%
WACC floor/cap	3.50/25.00%
Terminal reinvestment	23.81%

Interpretation

These are methodological assumptions, not filing-calibrated Barrick estimates. Terminal closure dominates the reported operating proxy. The legacy positive terminal increases the mean by USD 1.14–1.49 billion relative to signed closure.

The quarterly bridge multiplies simulated gold price less unit Cost of Sales, both in USD per ounce, by production in thousand ounces, then divides by 1,000 to obtain component margin in USD millions. Four quarterly margins are summed before annual taxes, reinvestment and discounting. This proxy is not reported EBIT or FCFF: production must be reconciled with sales and inventory, and depreciation, capital expenditure and working capital must enter without double counting. Fixed operating vectors exclude mine-level shocks and price-cost dependence.

7 OPERATING-PROXY SENSITIVITIES

Table 4. Audited Bates–Hawkes operating proxy.

Metric	USD billion or %
P10	1.03
Median (P50)	59.82
P90	172.60
Mean	77.95
Negative proxy paths	9.30%
Five-year-only median	12.89
Legacy floor mean uplift	1.49

Interpretation

These are unconstrained aggregate operating proxies, not enterprise or equity values. The Q-law/WACC combination has no demonstrated corporate pricing interpretation. Negative outcomes are not negative limited-liability share prices; no market comparison is made.

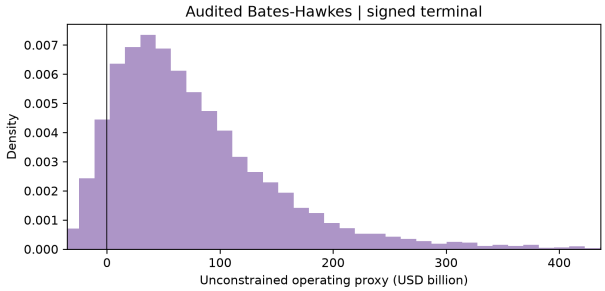


Figure 10. Signed-terminal operating proxy under audited Bates–Hawkes parameters. The common contract is reused across 8,192 paths. Display limits are the 0.5th–99.5th percentiles; the full tails remain in the data. This is not a confidence interval or physical probability law.

7.1 Cross-model central tendency

Audited parameter refinements dated 7 September use the unchanged 2 September market sample. Cross-model medians range from USD 59.12 to 60.10 billion and P90 from USD 172.60 to 174.02 billion. Their proximity should not be interpreted as a robust ranking of companies or models.

Table 5. Aggregate proxy medians, USD billion.

Gold engine	Median	Five-year only
Black–Scholes	60.10	13.17
Heston	59.77	13.08
Bates–Poisson	59.12	12.83
Full Bates–Hawkes	59.82	12.89

The simulation spread describes conditional scenario dispersion; Monte Carlo error concerns numerical estimates. Doubling paths to 16,384, doubling fine steps to 520 and changing the seed gives Bates–Hawkes medians of 59.34, 61.59 and 61.38 billion versus 59.82 at baseline. Such numerical variation exceeds some cross-model median gaps. Negative values expose the unconstrained mapping, not negative limited-liability shares. The data retain these outcomes rather than silently flooring them.

7.2 Structural illustration, not model selection

Bates–Hawkes illustrates the richest law, not a preferred corporate valuation. Its audited branching ratio is 0.5594 and kernel half-life $\log(2)/\beta = 0.2522$ years. These are risk-neutral fitted parameters, not measured geopolitical persistence. The earlier OOS exercise below uses the original, unrefined origin fits.

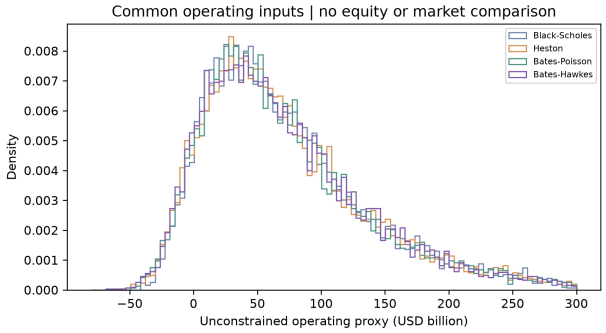


Figure 11. Audited operating-proxy distributions under four gold-price laws. Operations, WACC and signed terminal closure are common. No share count, equity bridge or market reference is used.

8 COPPER EXTENSION: HISTORY AND PAIRED OPERATING VALUES

8.1 Perimeter and units

The original gold-only study omitted copper. The extension admits three producing mines: Lumwana (100%), Zaldívar (50%) and Jabal Sayid (50%). Their Q2 2026 attributable sales are 40, 9 and 5 kt, and Cost of Sales is 3.14, 4.89 and 2.63 USD/lb, respectively (Barrick, 2026b). The sales quantities already incorporate ownership; multiplying by those shares again would understate the two joint ventures. Lumwana is consolidated, while the other two interests are equity-accounted (Barrick, 2025). Production and sales are distinct quantities; the copper margin uses sales.

With s_i in kt and P_t, c_i in USD/lb, quarterly attributable component margin in USD millions is

$$CM_t^{Cu} = \frac{2204.6226218488}{1000} \sum_i s_i(P_t - c_i). \tag{21}$$

Q2 sales and costs are held fixed for 20 quarters. This is a controlled operating scenario, not a mine-production forecast. CoS includes depreciation; reported capital expenditure is not separately deducted in this proxy. Expansion at Lumwana, Reko Diq, other gold assets, reserves and closure obligations remain outside the model.

8.2 Historical surface and exercise treatment

The IB API collection selected ten historical dates between 9 July and 2 September 2026 before scoring. Each date used 112 uniformly spaced out-of-the-money contract locations, rather than thousands of quotes. There are 912 returned quote rows across the pilot; missing responses are not fabricated. The common admission rule requires at least 64 eligible observations and three expiries, option and matched-future quote ages no greater than 120 seconds, and a relative option spread no greater than 20%. Seven dates survive, creating six origin–target pairs with gaps of 11, 23, 9, 10, 1 and 1 calendar days. July 10, July 31 and August 3 fail the quality gate. Historical data therefore exist, but this is a sparse pilot rather than a daily panel; the other gold reference dates were not collected in this experiment.

Copper is quoted in USD/lb through COMEX HG futures and their American options. A 600-step constant-volatility CRR inversion removes the early-exercise convention before fitting European forward-price engines; a 300-step comparison checks discretization. This transformation does not solve early exercise under the full stochastic-volatility/jump models. Each option is matched to its own underlying future; Black–76 is the constant-volatility benchmark. Origin calibration uses the dated Treasury curve, with no target-date quote entering parameter selection. The 2 September current fit uses all 86 eligible observations.

8.3 Temporal tests

Origin parameters are frozen and their latent states projected over the actual calendar gap. Target forward prices and rates condition repricing; the test is not an unconditional physical-price forecast. All four models share 480 target observations. Persistence uses origin IV interpolation inside the maturity–log-moneyness convex hull and has 375 supported targets. Equal-date scoring prevents dense dates from dominating; unsupported persistence points are omitted from its comparison.

Table 6. Copper rolling temporal repricing, six date pairs. RMSE is the mean daily RMSE in IV basis points; R^2_p is against observed-IV persistence on common support.

Model	RMSE (bp)	R^2_p
Black–76	106.48	−1.294
Heston	71.46	−0.422
Bates	70.49	−0.389
Bates–Hawkes	67.65	−0.208

No model beats persistence in the rolling experiment. As a robustness exercise, fixing 9 July as the sole calibration origin gives persistence R^2 of −0.345, −0.282, 0.193 and 0.182 in the same model order, on 333 supported targets. The positive Bates and Hawkes point estimates are conditional on this different origin, gap and support; they do not overturn the rolling result or establish significance with six dependent target dates.

8.4 Paired five-year contribution

Under the artificial law $\mathcal{R}$, the price scenario uses log-linear matched-future anchors at actual delivery dates, flat beyond the final observed delivery at 0.728 years. This schedule is neither a fixed-futures drift nor a fitted physical expectation. Five years therefore requires substantial extrapolation. The Paper's revised gold parameters, operating vectors and 8,192 WACC paths are reused exactly for gold-only and combined runs. The primary case assumes independent commodity paths and no terminal value. Whole-path permutations with terminal Gaussian-score correlation ±0.5 and a signed perpetuity are separate sensitivities, not estimated price dependence or reserve-supported continuation.

Table 7. Paired five-year means in USD bn. Δ is the mean pathwise copper increment; SE is its Monte Carlo standard error in USD mn. These are operating proxies.

Model	Gold	Gold+Cu	Δ	SE
Black benchmark	14.857	17.484	2.627	23.0
Heston	14.847	17.465	2.618	39.4
Bates	14.799	17.466	2.666	40.4
Bates–Hawkes	14.795	17.438	2.643	37.0

The additional margin enters the same tax and growth–ROIC operator before discounting. The copper increment is approximately USD 2.62–2.67 bn under this contract; it cannot be added to the gold signed-terminal medians reported earlier, which use a different statistic and horizon. Monte Carlo standard errors measure simulation noise only. Mine-level forecast, commodity basis, accounting and parameter uncertainty remain outside them. The extension closes an omitted operating component while leaving the corporate FCFF and equity bridges open.

9 OUT-OF-SAMPLE EVIDENCE AND MODEL RISK

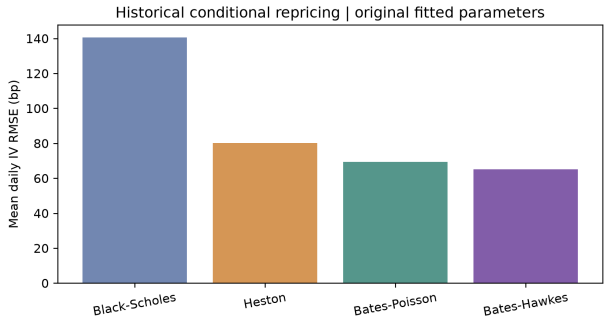


Figure 12. Dense-only rolling evidence on 30 forecast origins and 4,667 common target observations. The original fits give Bates–Hawkes the lowest mean daily IV RMSE.

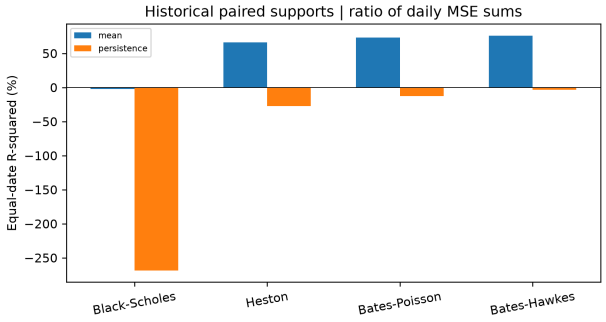


Figure 12 (continued). Positive R^2_{OOS} indicates improvement over the stated benchmark.

9.1 Conditional next-date repricing

Model comparisons use 4,667 targets; persistence comparisons use the 4,026 inside the origin interpolation domain. Benchmark scores therefore have different supports; equal date weighting limits the influence of densely populated dates.

Reported RMSE is the mean of daily RMSEs, not the root of mean daily MSE used by R^2. Values are 140.6528 bp for Black–Scholes, 80.1964 bp for Heston, 69.3055 bp for Bates and 65.0507 bp for Full Bates–Hawkes. Relative to the origin mean-IV benchmark, the corresponding R^2_{OOS} values are −2.58%, 66.07%, 72.97% and 75.81%. None beats interpolated IV-surface persistence, but Full Bates–Hawkes is closest at −3.68%, followed by Bates at −12.92% and Heston at −27.24%.

Model-risk conclusion

Historical OOS uses the original fits; audited current fits are a separate experiment. A paired five-date block bootstrap gives a Hawkes-minus-Bates MSE interval $[-8.95, -2.77] \times 10^{-6}$, exploratory with only 30 dates.

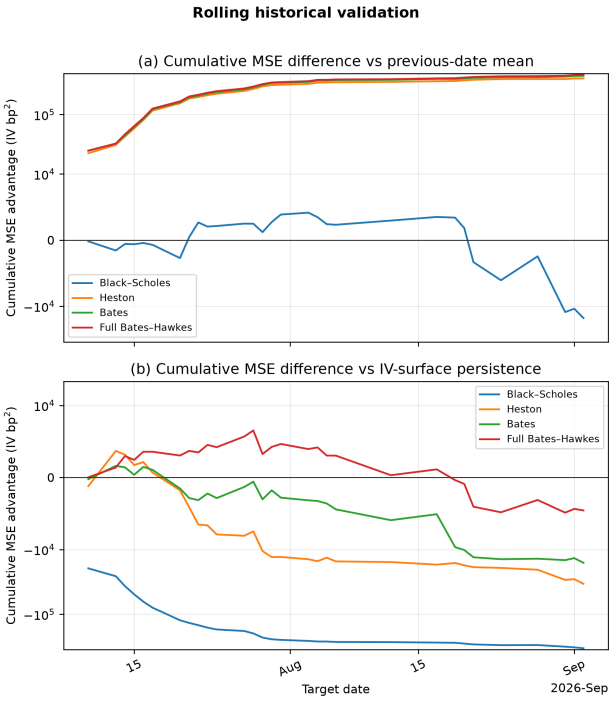


Figure 13. Cumulative rolling MSE differences against the origin mean IV and interpolated IV-surface persistence. Upward movement favours the option model; the stronger persistence benchmark remains difficult to beat.

9.2 Jump-layer interpretation

The self-exciting extension is motivated by clustered option-implied jump activity. Relative to the constant Poisson baseline, the Hawkes specification permits intensity bursts followed by endogenous decay. The branching ratio and decay parameter have different meanings; fitted risk-neutral self-excitation is not evidence of physical news clustering.

Without a dated news filtration or parameter-stability evidence, these option-implied bursts cannot be identified as causal responses to

geopolitical events.

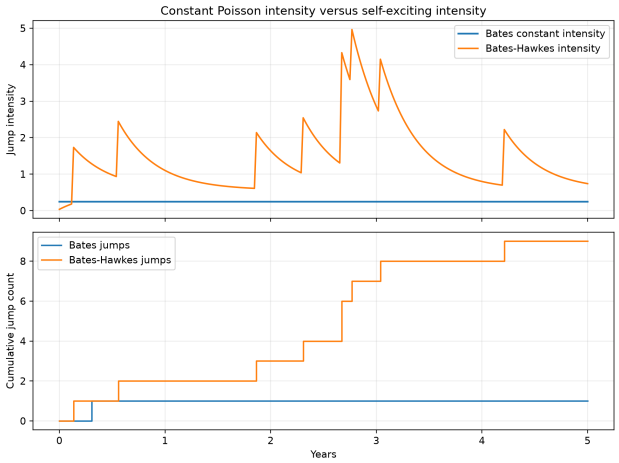


Figure 14. Illustrative simulated Poisson and Hawkes paths under the original parameter set. A single realization cannot establish clustering in observed data.

9.3 Interpretive limitations and validation

Reproducibility, numerical accuracy and economic validity are distinct. A good option fit cannot validate the GLD-to-gold mapping or the Q-law/WACC combination. The experiment identifies sensitivity to a chosen numerical contract, not corporate fair value.

The old run equated enterprise and equity proxies using zero bridge placeholders. This revision removes that share-price interpretation; debt, cash and other claims remain unreconciled. The new copper operating extension does not close this corporate bridge. A genuine valuation must start from one dated filing and the matching listed security, establish ownership and attributable production, and distinguish component margin from reported operating profit. Depreciation, cash taxes, working capital, sustaining capital expenditure and growth investment must then enter consistently, with no deduction counted twice. Net debt, non-operating assets, streams and other claims must be matched to a diluted share count on the same basis. Only then can a per-share output acquire a filing-reconciled interpretation.

9.4 Operating uncertainty and research priorities

Production, grade, recovery, mine mix and costs may depend on gold prices, inflation and exchange rates. Independence is not automatic: operating dependence and Goodman variance propagation require explicit assumptions. Technology scenarios must include implementation costs and investment; a smooth conditional cost mean does not imply absence of uncertainty.

A mining-company terminal block needs an explicit resource interpretation. Continued production may require reserve conversion, replacement investment or new projects, while depletion and closure liabilities can reduce the duration and value of the cash-flow stream. A positive WACC-growth spread controls a mathematical denominator but does not establish that a perpetual operating horizon is economically justified. Finite-resource and going-concern closures should be compared under coherent investment assumptions before attributing differences entirely to the gold-price model. Such a comparison is a future validation exercise, not a result of the present run.

Market maturities end at 653 days, well before five scenario years and perpetuity. Fixed parameters beyond that horizon are an extrapolation assumption, not market-identified information. The five-year-only comparison removes terminal continuation but does not calibrate depletion, replacement investment or closure costs. A bucketed term-structure and finite-reserve model remain necessary for an economic valuation.

CONCLUSION

Holding operations and corporate assumptions fixed isolates the effect of gold-price model choice. Nested-candidate safeguards improve the current Bates and Hawkes fits. Historical option-surface OOS remains distinct and no original engine beats interpolated persistence. Numerical and terminal sensitivity prevent a corporate value ranking.

The copper extension admits three producing mines and adds a conditional five-year operating contribution. Historical snapshots support actual temporal testing, but the sparse rolling experiment does not beat IV persistence. Its positive fixed-origin Bates and Hawkes comparisons are exploratory. The combined output remains an attributable component-margin sensitivity, without a reconciled FCFF or equity bridge.

The persistent-cost assumption is prudential only relative to otherwise identical efficiency-improvement scenarios, not a lower bound. The contribution is an auditable interdisciplinary framework, not a certified value: commodity basis, probability measure, operating uncertainty, corporate accounting and finite-resource closure remain the priorities for economic validation.

The next required result is a validated physical commodity forecast for each admitted model, with an explicit link to option calibration and Barrick's realised prices. This is a prerequisite for the intended DCF, not an optional robustness check. Until price, operating and accounting gates are met, the current calculations remain conditional sensitivities.

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